

**Nick's Question 0.1** In the solution (1.8)

$$u(t, x) = J_0(t, x) + I(t, x)$$

Why isn't the  $I_t^\gamma$  operator represented? It seems like (1.8) is the same whether  $I_t^\gamma$  is there or not.

**Solution:** It is built into the triplet in equation (1.5)

**Nick's Question 0.2 Proof of Theorem 3.3**

Why is (3.5) due to (3.1) and (3.3). I see why (3.1) is important but why (3.3)? Does  $\bar{\mathcal{K}}$  satisfy the sup property? This may be Theorem 3.4?

**Nick's Question 0.3 Theorem 3.5** What happens to the  $\hat{C}_t$  when we are finding the upper bound for  $\bar{m}_p$ . It seems like the

$$\frac{1}{t} \log \hat{C}_t$$

terms are being absorbed into the constant as  $t \rightarrow \infty$ . Is this true because of

$$\frac{1}{t} \log(t) \rightarrow 0 \quad t \rightarrow \infty?$$

**Nick's Question 0.4 possible typo on page 5087 equation 4.16 -4.18** I think that in the asymptotic bound for  $Y$ , that  $\gamma$  needs to be replaced with  $1 - \beta$  not just 1 for the  $Z^*$  case. This can be seen from (4.4)-(4.6)

**Nick's Question 0.5**

1. Where does the  $\int_0^t d\tau$  come from and why can we remove the laplace transform from  $f$  in the proof of theorem 4.1. I see using

$$\mathcal{L}[t^{\beta-1} E_{\alpha,\beta}(-\lambda t^\alpha)](s) = \frac{s^{\alpha-\beta}}{s^\alpha + \lambda}$$

then

$$\frac{s^{-\gamma}}{s^\beta + \frac{\nu}{2} |\xi|^\alpha} = \mathcal{L} \left[ t^{\beta+\gamma-1} E_{\beta,\beta+\gamma} \left( -\frac{\nu}{2} |\xi|^\alpha t^\beta \right) \right] (s)$$

so from the right term from the top of page 5090 we have

$$\mathcal{L} \left[ t^{\beta+\gamma-1} E_{\beta,\beta+\gamma} \left( -\frac{\nu}{2} |\xi|^\alpha t^\beta \right) \right] (s) \tilde{f}(s, \xi)$$

but I do not know why we can use this to get the  $\int_0^t d\tau$  term in the expression for  $\hat{u}(t, \xi)$ .

2. In order to show that  $Y(t, x) = {}_t D_+^\theta Z(t, x)$  you showed that

$$\mathcal{F}(Y(t, \cdot))(\xi) = {}_t D_+^\theta \mathcal{F}(Z(t, \cdot))(\xi)$$

Does the  ${}_tD_+^\theta$  operator commute with the Fourier transform?